100 Year MovieTrend Analysis

Objective / Problem Statement:

- To explore factors influencing movie success by analyzing relationships between budget, revenue, genre, runtime, ratings, and popularity.

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1. Dataset Overview

Dataset Name: **Movie Dataset**
Source: Kaggle

Number of Rows: 4803
Number of Columns: 19

Column Name	Description
index	Unique row number or identifier for each record.
budget	Total production budget of the movie (usually in USD).
genres	Movie category or type (e.g., Action, Drama, Comedy).
homepage	Official website URL of the movie.
id	Unique movie ID used in the dataset or database.
keywords	Important tags or keywords related to the movie content or theme.
original_language	Original language in which the movie was produced.
original_title	Original name of the movie in its native language.
overview	Short summary or description of the movie storyline.
popularity	Popularity score based on user activity, views, or engagement metrics.
production_companies	Companies involved in producing the movie.

production_countries	Countries where the movie was produced.
release_date	Official release date of the movie.
revenue	Total earnings or box office revenue generated by the movie.
runtime	Duration of the movie (in minutes).
spoken_languages	Languages spoken in the movie.
status	Release status (e.g., Released, Post Production).
tagline	Promotional slogan or tagline of the movie.
title	Display or main title of the movie.
vote_average	Average rating given by users/viewers.
vote_count	Total number of votes or ratings received.
cast	List of actors or performers in the movie.
crew	List of crew members (director, producer, writer, etc.).
director	Name of the movie director.

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2. Data Cleaning (Python)

✅ Handling missing values
✅ Removing duplicates
✅ Correcting data types
✅ Standardizing text values
✅ Treating outliers

[Python File]

Why cleaning was necessary:

- Improve data accuracy

- Avoid analysis errors
- Prepare dataset for modeling

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3. SQL Analysis

Business Question 1:

How does movie popularity change based on original language, and which languages show the highest average performance?

SQL Query:

```
SELECT
    original_language,
    ROUND(AVG(revenue), 1) AS avg_rev,
    ROUND(AVG(popularity), 1) AS avg_pop,
    CASE
        WHEN AVG(revenue) > 25356448 THEN 'high'
        WHEN AVG(revenue) > 1000000 THEN 'medium'
        ELSE 'low'
    END AS avg_rev_compare,
    CASE
        WHEN AVG(popularity) > 20 THEN 'high'
        WHEN AVG(popularity) > 10 THEN 'medium'
        ELSE 'low'
    END AS avg_pop_compare
FROM
    Movies_data
GROUP BY original_language
ORDER BY avg_rev DESC;
```

Output Screenshot:

original_language	avg_rev	avg_pop	avg_rev_compare	avg_pop_compare
te	100000000.0	16.3	high	medium
en	86478849.7	22.2	high	high
ja	66028915.6	25.7	high	high
xx	55260558.0	6.5	high	low
zh	41734979.7	9	high	low
cn	33740155.8	10.6	high	medium
da	29898888.0	17.7	high	medium

Insight:

Movies in most languages generate high revenue, but only English and Japanese films combine both high revenue and high popularity, showing stronger overall performance.

Business Question 2:

Identify movies that performed significantly better or worse than the average revenue within their genre

SQL Query:

```
WITH genre_avg AS (
   SELECT
      genres,
      AVG(revenue) AS avg_genre_revenue
   FROM Movies_data
   GROUP BY genres
)

SELECT
   m.genres,
```

```
    g.avg_genre_revenue,
    m.original_title,
    m.revenue,
    CASE
        WHEN m.revenue > g.avg_genre_revenue THEN 'Above Average'
        ELSE 'Below Average'
    END AS revenue_performance
FROM Movies_data AS m
JOIN genre_avg AS g
    ON m.genres = g.genres
ORDER BY
    m.genres,
    m.revenue DESC;
```

Output Screenshot:

genres	avg_rev	original_title	revenue	compare
Action Adventure Fantasy Science Fiction	781703238.1667	Beastmaster 2: Through the Portal of Time	869325	bad
Action Adventure Fantasy Science Fiction	781703238.1667	Superman II	190458706	bad
Action Adventure Fantasy Science Fiction	781703238.1667	Superman	300218018	bad
Action Adventure Fantasy Science Fiction	781703238.1667	X-Men: Days of Future Past	747862775	bad
Action Adventure Fantasy Science Fiction	781703238.1667	Man of Steel	662845518	bad
Action Adventure Fantasy Science Fiction	781703238.1667	Avatar	2787965087	good
Adventure Fantasy Action	580701593.9231	Red Sonja	6948633	bad
Adventure Fantasy Action	580701593.9231	Conan the Destroyer	100000000	bad
Adventure Fantasy Action	580701593.9231	Conan the Barbarian	79114085	bad
Adventure Fantasy Action	580701593.9231	The Lord of the Rings: The Two Towers	926287400	good

Insight:

Only a few movies (like *Avatar* and *The Lord of the Rings: The Two Towers*) perform above their genre's average revenue, showing that most films earn below the genre benchmark while a small number of blockbusters drive overall averages higher.

Business Question 3:

Is there a relationship between movie runtime and audience ratings or revenue performance?

SQL Query:

```
SELECT
   CASE
      WHEN runtime < 90 THEN 'short'
      WHEN runtime BETWEEN 90 AND 120 THEN 'medium'
      ELSE 'lagre'
   END AS runtime_performance,
   COUNT(*) AS number_of_movies,
   ROUND(AVG(revenue), 1)
FROM
   Movies_data
GROUP BY runtime_performance;
```

Output Screenshot:

runtime_performance	number_of_movies	ROUND(AVG(revenue), 1)
lagre	988	165254394.8
medium	3110	65737090.9
short	705	38842769.6

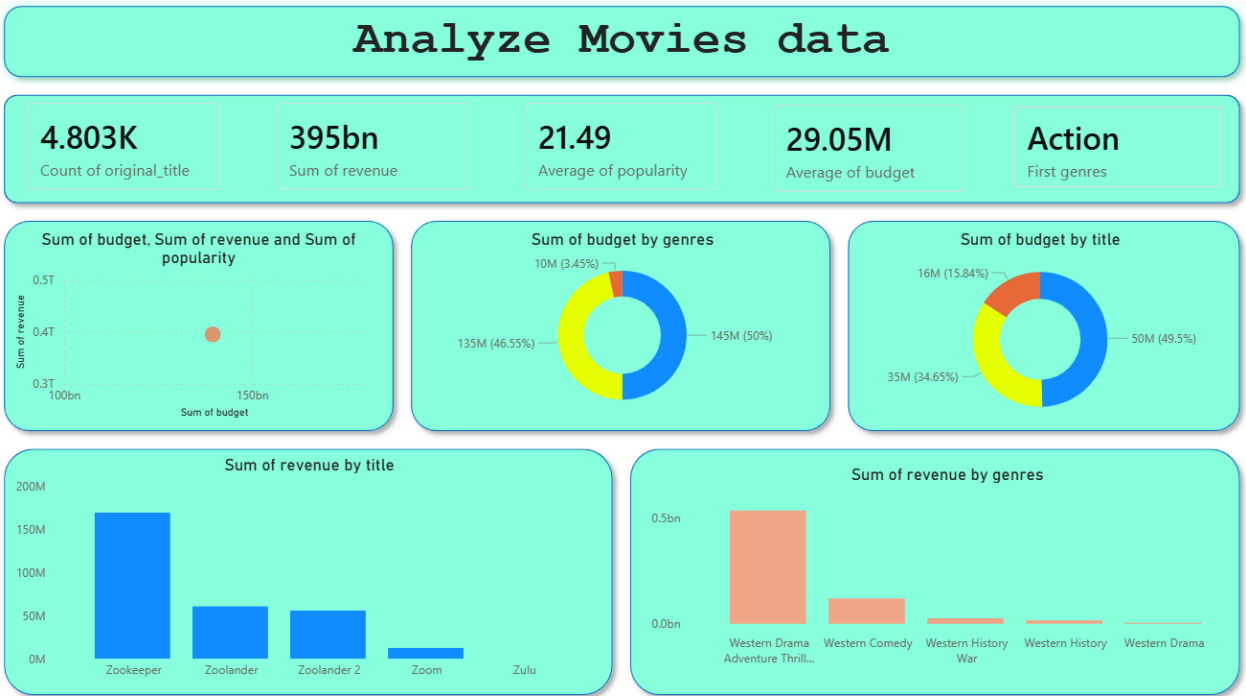
Insight:

Longer runtime movies generate significantly higher average revenue than medium and short films, suggesting that big-budget or blockbuster films tend to have longer durations.

[All Business Question]

[Output of Question]

4. Dashboard Summary (Power BI / Tableau)



Explain:

- Main KPIs

1. Shows total number of movies (4.803K), giving a quick overview of dataset size and analysis coverage.
2. Total revenue (395bn) indicates overall financial performance of all movies combined.
3. Average popularity (21.49) reflects audience engagement and general movie interest levels.
4. Average budget (29.05M) helps understand typical investment made per movie.
5. First genre (Action) highlights the primary or most selected genre in analysis.

- Filters used
- Shows total number of movies (4.803K), giving a quick overview of dataset size and analysis coverage.
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- Key visuals
- Scatter plot shows relationship between budget, revenue, and popularity to identify high-performing movies.
- Donut charts display budget distribution by genre and title for quick comparison.
- Bar chart of revenue by title highlights top-performing movies financially.
- Revenue by genre chart compares financial success across different categories.
- KPI cards provide quick summary metrics for faster understanding of overall insights.

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5. Key Insights

1️⃣ Longer runtime movies have the highest average revenue, indicating blockbuster-style productions.

2️⃣ Medium-length movies are most common but generate moderate revenue compared to longer films.

3️⃣ Short movies show the lowest revenue performance overall.

4️⃣ Only a few movies perform above their genre average, showing revenue is driven by top hits.

5️⃣ Some genres rely heavily on blockbuster films to increase average revenue.

6️⃣ English and Japanese movies combine high revenue with high popularity, indicating strong global appeal.

7️⃣ Certain languages achieve high revenue despite lower popularity, suggesting strong regional markets.

8️⃣ Genre benchmarks help identify whether a movie performs above or below expectations.

9️⃣ Popularity alone does not guarantee higher revenue performance.

🔟 Overall, the movie industry shows a “hit-driven” pattern where a small number of high-performing films drive most financial success.

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6. Recommendations

Business suggestions based on analysis:

- Invest in high-performing genres that consistently generate above-average revenue to maximize ROI.
- Prioritize longer runtime or blockbuster-style productions where data shows higher earning potential.
- Collaborate with directors or production teams linked to high-performing films to increase success probability.
- Strengthen marketing strategies for movies with high revenue potential but lower popularity to improve audience reach.
- Analyze regional/language markets separately and expand successful local content globally.
- Reduce investment in consistently underperforming segments or redesign their strategy.
- Use genre benchmarks to evaluate project performance before and after release.
- Leverage data-driven decision-making for future movie planning and distribution strategies.

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7. Conclusion

Short summary of:

- What you analyzed

  Movie dataset using Python and SQL to study relationships between genres, runtime, language, popularity, and revenue performance.

- What you discovered

  Longer movies tend to generate higher revenue, only a few blockbuster films perform above genre averages, and some languages combine strong popularity with high earnings while others rely on regional success.

- Business value

  Helps studios understand which movie characteristics drive financial success, improve investment decisions, optimize marketing strategies, and identify high-performing segments for future production.